**CSE370 : Database Systems**

**Assignment 01 | Spring 2026**

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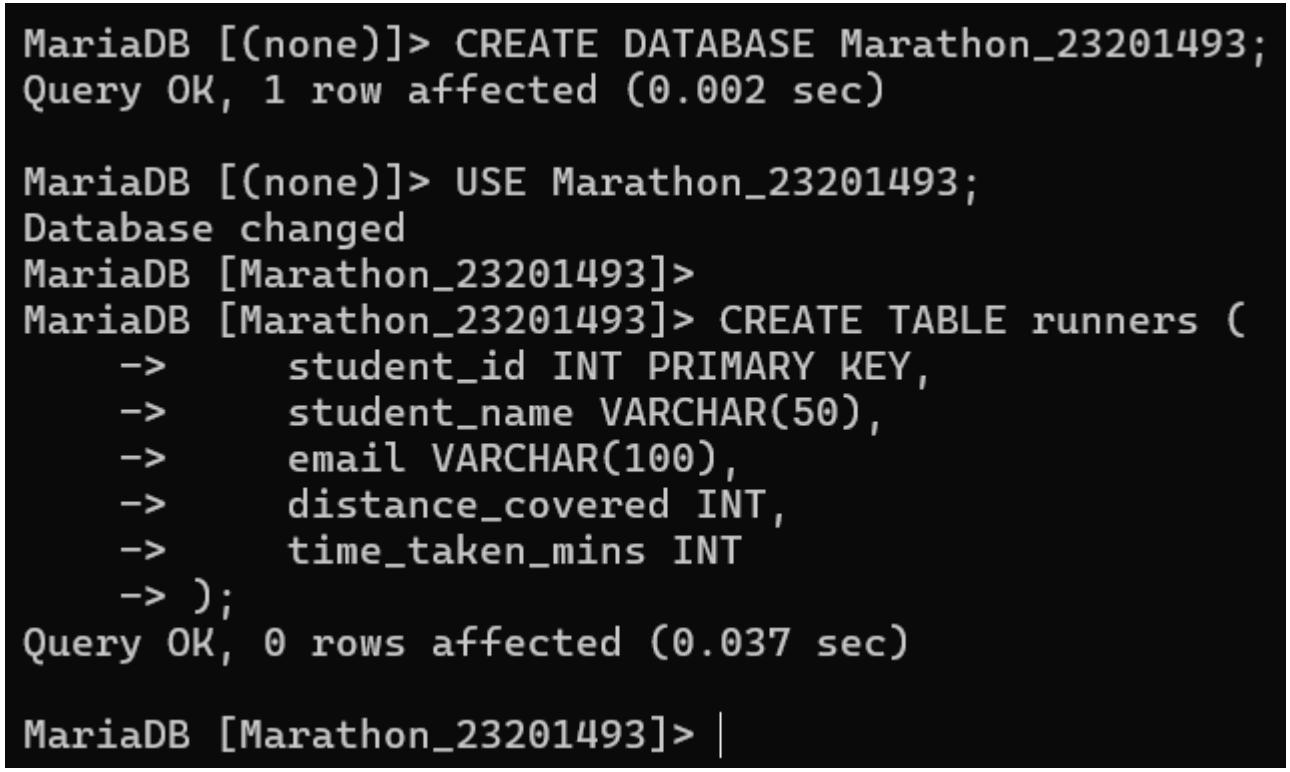
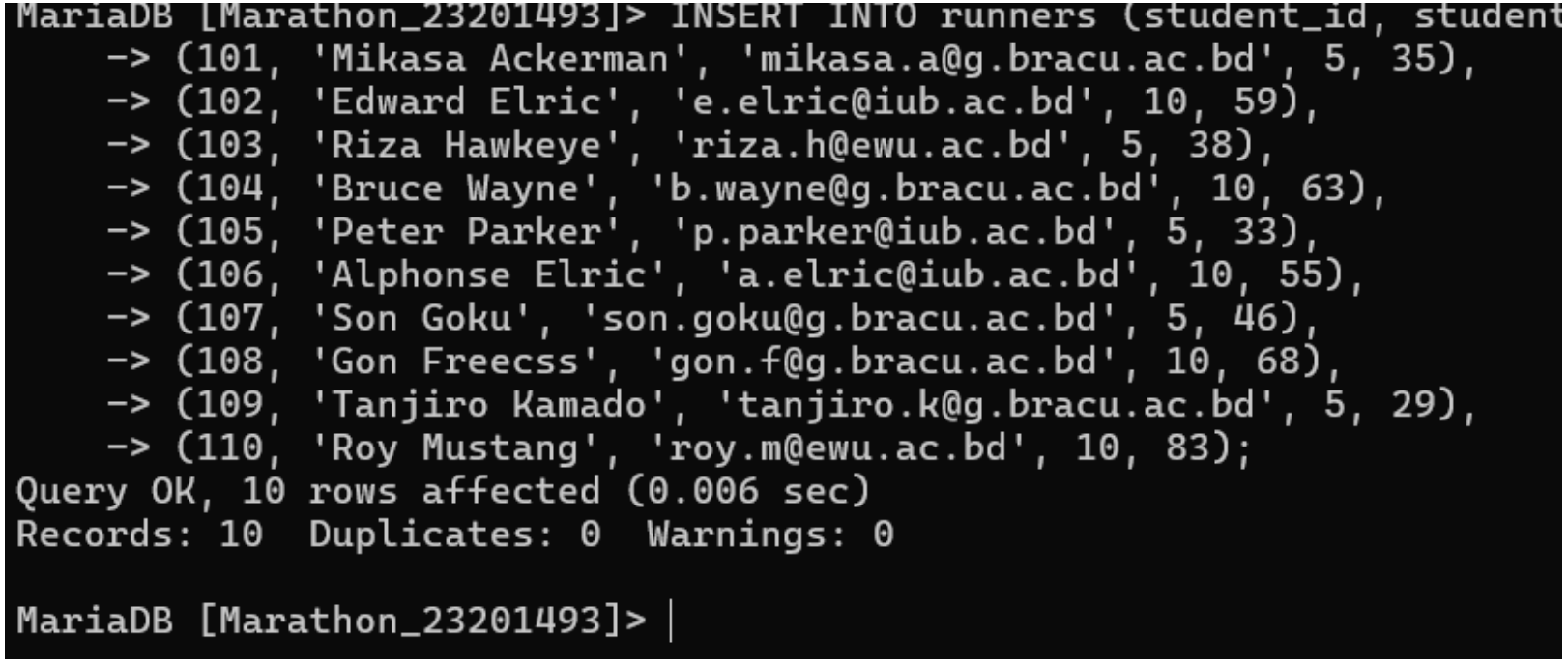
| | |
|---|---|
| **No 1 Query (as Plain Text)** | CREATE TABLE runners (<br>student_id INT PRIMARY KEY,<br>student_name VARCHAR(50),<br>email VARCHAR(100),<br>distance_covered INT,<br>time_taken_mins INT<br>); |
| **No 1 SS (of Query & Output in Shell)** |  |

```
MariaDB [(none)]> CREATE DATABASE Marathon_23201493;
Query OK, 1 row affected (0.002 sec)

MariaDB [(none)]> USE Marathon_23201493;
Database changed
MariaDB [Marathon_23201493]>
MariaDB [Marathon_23201493]> CREATE TABLE runners (
    ->     student_id INT PRIMARY KEY,
    ->     student_name VARCHAR(50),
    ->     email VARCHAR(100),
    ->     distance_covered INT,
    ->     time_taken_mins INT
    -> );
Query OK, 0 rows affected (0.037 sec)

MariaDB [Marathon_23201493]>
```

| | |
|---|---|
| **No 2 Query (as Plain Text)** | INSERT INTO runners (student_id, student_name, email, distance_covered, time_taken_mins) VALUES<br>(101, 'Mikasa Ackerman', 'mikasa.a@g.bracu.ac.bd', 5, 35),<br>(102, 'Edward Elric', 'e.elric@iub.ac.bd', 10, 59),<br>(103, 'Riza Hawkeye', 'riza.h@ewu.ac.bd', 5, 38),<br>(104, 'Bruce Wayne', 'b.wayne@g.bracu.ac.bd', 10, 63),<br>(105, 'Peter Parker', 'p.parker@iub.ac.bd', 5, 33),<br>(106, 'Alphonse Elric', 'a.elric@iub.ac.bd', 10, 55),<br>(107, 'Son Goku', 'son.goku@g.bracu.ac.bd', 5, 46),<br>(108, 'Gon Freecss', 'gon.f@g.bracu.ac.bd', 10, 68),<br>(109, 'Tanjiro Kamado', 'tanjiro.k@g.bracu.ac.bd', 5, 29),<br>(110, 'Roy Mustang', 'roy.m@ewu.ac.bd', 10, 83); |
| **No 2 SS (of Query & Output in Shell)** |  |

```
MariaDB [Marathon_23201493]> INSERT INTO runners (student_id, student
    -> (101, 'Mikasa Ackerman', 'mikasa.a@g.bracu.ac.bd', 5, 35),
    -> (102, 'Edward Elric', 'e.elric@iub.ac.bd', 10, 59),
    -> (103, 'Riza Hawkeye', 'riza.h@ewu.ac.bd', 5, 38),
    -> (104, 'Bruce Wayne', 'b.wayne@g.bracu.ac.bd', 10, 63),
    -> (105, 'Peter Parker', 'p.parker@iub.ac.bd', 5, 33),
    -> (106, 'Alphonse Elric', 'a.elric@iub.ac.bd', 10, 55),
    -> (107, 'Son Goku', 'son.goku@g.bracu.ac.bd', 5, 46),
    -> (108, 'Gon Freecss', 'gon.f@g.bracu.ac.bd', 10, 68),
    -> (109, 'Tanjiro Kamado', 'tanjiro.k@g.bracu.ac.bd', 5, 29),
    -> (110, 'Roy Mustang', 'roy.m@ewu.ac.bd', 10, 83);
Query OK, 10 rows affected (0.006 sec)
Records: 10  Duplicates: 0  Warnings: 0

MariaDB [Marathon_23201493]>
```

| | |
|---|---|
| | |
| **No 3 Query (as Plain Text)** | SELECT * FROM runners<br>WHERE distance_covered > 6; |
| **No 3 SS (of Query & Output in Shell)** | |

```
MariaDB [Marathon_23201493]> SELECT * FROM runners
    -> WHERE distance_covered > 6;
+------------+----------------+-----------------------+------------------+-----------------+
| student_id | student_name   | email                 | distance_covered | time_taken_mins |
+------------+----------------+-----------------------+------------------+-----------------+
|        102 | Edward Elric   | e.elric@iub.ac.bd     |               10 |              59 |
|        104 | Bruce Wayne    | b.wayne@g.bracu.ac.bd |               10 |              63 |
|        106 | Alphonse Elric | a.elric@iub.ac.bd     |               10 |              55 |
|        108 | Gon Freecss    | gon.f@g.bracu.ac.bd   |               10 |              68 |
|        110 | Roy Mustang    | roy.m@ewu.ac.bd       |               10 |              83 |
+------------+----------------+-----------------------+------------------+-----------------+
5 rows in set (0.001 sec)

MariaDB [Marathon_23201493]>
```

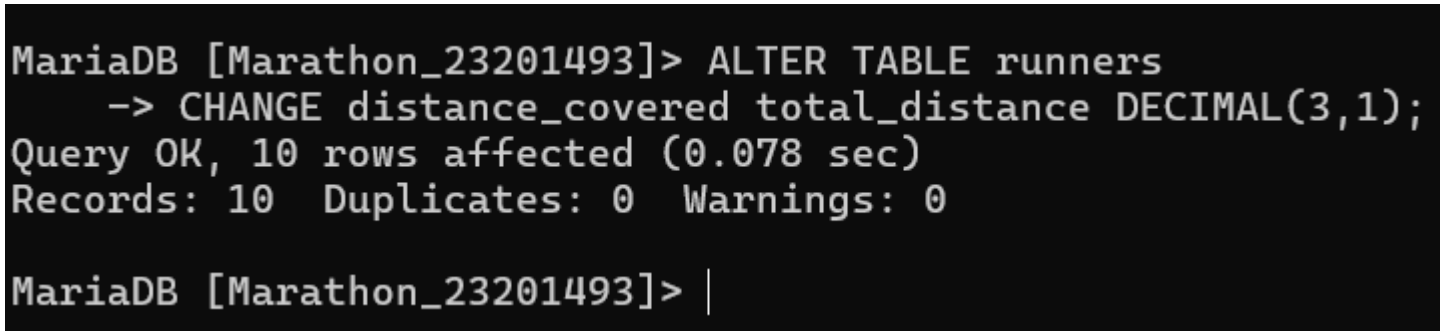
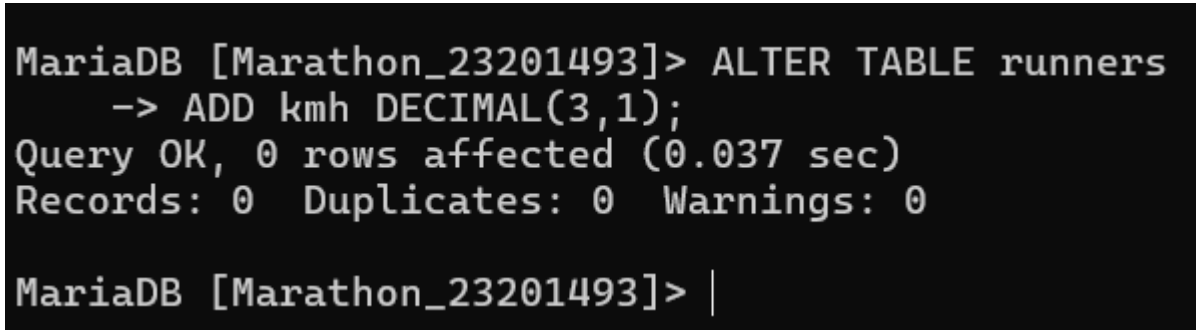
| | |
|---|---|
| | |
| **No 4 Query (as Plain Text)** | SELECT UPPER(student_name) AS student_name<br>FROM runners<br>ORDER BY time_taken_mins DESC; |

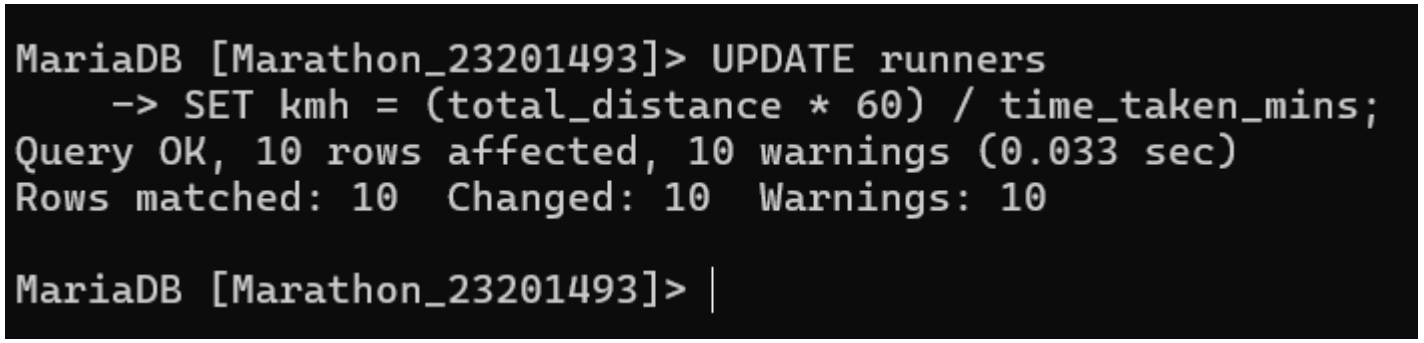
| | |
|---|---|
| **No 4 SS (of Query & Output in Shell)** | |

```
MariaDB [Marathon_23201493]> SELECT UPPER(student_name) AS student_name
    -> FROM runners
    -> ORDER BY time_taken_mins DESC;
+-----------------+
| student_name    |
+-----------------+
| ROY MUSTANG     |
| GON FREECSS     |
| BRUCE WAYNE     |
| EDWARD ELRIC    |
| ALPHONSE ELRIC  |
| SON GOKU        |
| RIZA HAWKEYE    |
| MIKASA ACKERMAN |
| PETER PARKER    |
| TANJIRO KAMADO  |
+-----------------+
10 rows in set (0.001 sec)

MariaDB [Marathon_23201493]>
```

| | |
|---|---|
| **No 5 Query (as Plain Text)** | ALTER TABLE runners<br>CHANGE distance_covered total_distance DECIMAL(3,1); |

| | |
|---|---|
| **No 5 SS (of Query & Output in Shell)** |  |
| | |
| **No 6 Query (as Plain Text)** | ALTER TABLE runners<br>ADD kmh DECIMAL(3,1); |
| **No 6 SS (of Query & Output in Shell)** |  |
| | |
| **No 7 Query (as Plain Text)** | UPDATE runners<br>SET kmh = (total_distance * 60) / time_taken_mins; |

| | |
|---|---|
| **No 7 SS (of Query & Output in Shell)** |  |
| | |
| **No 8 Query (as Plain Text)** | SELECT student_id, student_name, kmh,<br>(total_distance / 10) * (time_taken_mins / 60) * 100 AS points<br>FROM runners; |

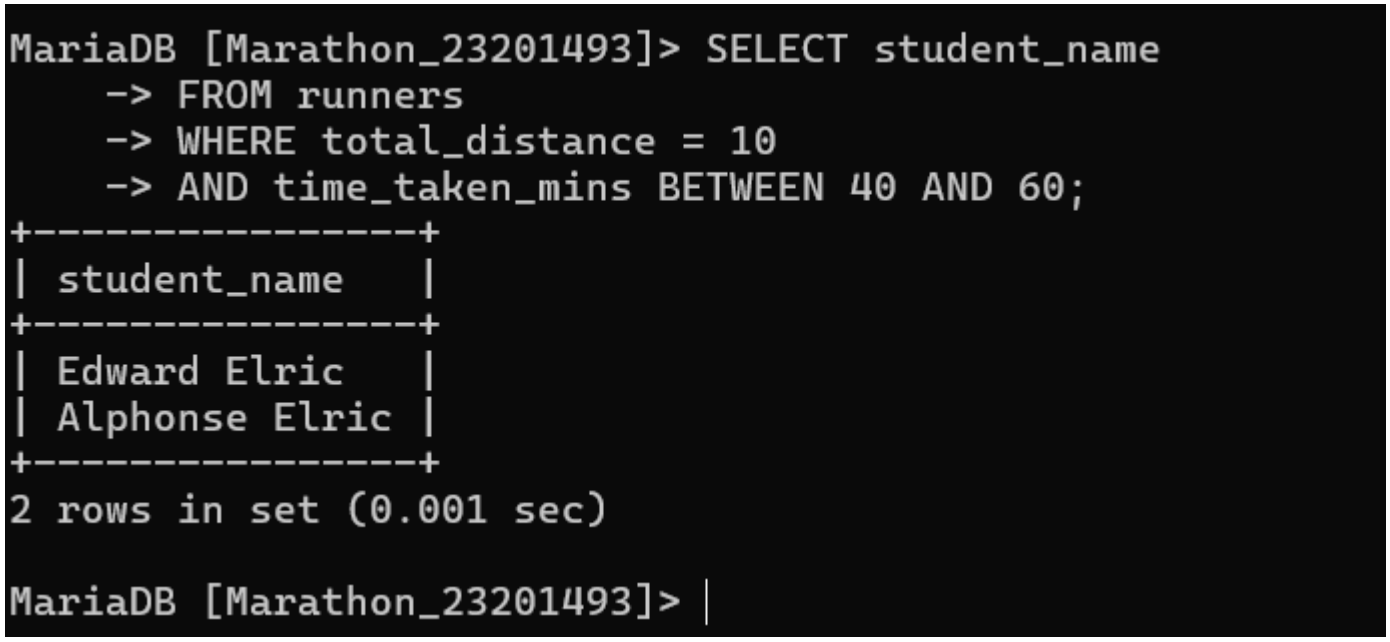
**No 8 SS**
**(of Query & Output in Shell)**

```
MariaDB [Marathon_23201493]> SELECT student_id, student_name, kmh,
    -> (total_distance / 10) * (time_taken_mins / 60) * 100 AS points
    -> FROM runners;
+------------+-----------------+------+---------------+
| student_id | student_name    | kmh  | points        |
+------------+-----------------+------+---------------+
|        101 | Mikasa Ackerman |  8.6 |  29.166666650 |
|        102 | Edward Elric    | 10.2 |  98.333333300 |
|        103 | Riza Hawkeye    |  7.9 |  31.666666650 |
|        104 | Bruce Wayne     |  9.5 | 105.000000000 |
|        105 | Peter Parker    |  9.1 |  27.500000000 |
|        106 | Alphonse Elric  | 10.9 |  91.666666600 |
|        107 | Son Goku        |  6.5 |  38.333333300 |
|        108 | Gon Freecss     |  8.8 | 113.333333300 |
|        109 | Tanjiro Kamado  | 10.3 |  24.166666650 |
|        110 | Roy Mustang     |  7.2 | 138.333333300 |
+------------+-----------------+------+---------------+
10 rows in set (0.001 sec)

MariaDB [Marathon_23201493]>
```

**No 9 Query**
**(as Plain Text)**

```
SELECT student_name
FROM runners
WHERE total_distance = 10
AND time_taken_mins BETWEEN 40 AND 60;
```

| | |
|---|---|
| **No 9 SS (of Query & Output in Shell)** |  |
| | |
| **No 10 Query (as Plain Text)** | SELECT student_name<br>FROM runners<br>WHERE email LIKE '%@g.bracu.ac.bd'; |

```
MariaDB [Marathon_23201493]> SELECT student_name
    -> FROM runners
    -> WHERE total_distance = 10
    -> AND time_taken_mins BETWEEN 40 AND 60;
+----------------+
| student_name   |
+----------------+
| Edward Elric   |
| Alphonse Elric |
+----------------+
2 rows in set (0.001 sec)

MariaDB [Marathon_23201493]>
```

| | |
|---|---|
| **No 10 SS (of Query & Output in Shell)** | |

```
MariaDB [Marathon_23201493]> SELECT student_name
    -> FROM runners
    -> WHERE email LIKE '%@g.bracu.ac.bd';
+-----------------+
| student_name    |
+-----------------+
| Mikasa Ackerman |
| Bruce Wayne     |
| Son Goku        |
| Gon Freecss     |
| Tanjiro Kamado  |
+-----------------+
5 rows in set (0.001 sec)

MariaDB [Marathon_23201493]> |
```